

DSA

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Quiz

Adarsh Singh
063

1. b. Data and a pointer to the next node
2. c. NULL
3. b. Self-referential structure
4. b. head == NULL
5. b. newnode->next = head; head = newnode;
6. b. temp = head; head = head->next; free(temp);
7. c. Travelling from tail to head
8. b. n
9. c. Copying the data of p->next into p, then deleting p->next
10. b. while (p != NULL)
11. b. p->next == NULL
12. d. ~~Two pointers~~ Marking each visited node with flag
13. c. ~~Two~~ Three
14. b. Exactly one node
15. b. p = (struct node*) malloc(sizeof(struct node));
16. b. Avoid special-case handling for insertion and deletion head
17. c. The first node
18. ~~Two~~
19. b. loss of access to all node, causing a memory leak
20. ~~Two~~ c. Recursion applied to the tail
21. c. The last node
22. ~~Two~~

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23. struct node {
 int data;
 struct node *next
};

24. Output :- 30, 50, 70.

25. `struct node* reverseList(struct node* head)`
`{`

```

    struct node* prev = NULL;
    struct node* curr = head;
    struct node* next = NULL;
    while (curr != NULL) {
        next = curr->next;
        curr->next = prev;
        prev = curr;
        curr = next;
    }
    return prev;
}

```

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28. $5 \rightarrow 8 \rightarrow 12 \rightarrow 3 \rightarrow \text{NULL}$
 $7 \rightarrow 5 \rightarrow 8 \rightarrow 12 \rightarrow 3 \rightarrow \text{NULL}$
 $7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 9 \rightarrow \text{NULL}$
 $7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 9 \rightarrow \text{NULL}$

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